

Junran Wang

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EDUCATION

2025 - Present	Georgia Institute of Technology Master of Science in <i>Electrical and Computer Engineering</i>	Atlanta, GA GPA: 4.0/4.0
2021 - 2025	Tsinghua University Bachelor of Engineering in <i>Vehicle Engineering (Electronic Information)</i>	Beijing, China GPA: 4.0/4.0
2024	University at Buffalo, the State University of New York Visiting Scholar in <i>Department of Computer Science and Engineering</i>	Buffalo, NY

RESEARCH EXPERIENCE

ImmersedPrivacy: Evaluating VLMs' Privacy Awareness in the Physical World Jan 2026 - May 2026
Georgia Institute of Technology | First Author | **NeurIPS 2026** |  Website Advisor: Dr. Pan Li

- Built a **reusable data generation and evaluation pipeline** in Unity for agent privacy, producing **400+ interactive audio-visual scenarios across 40+ physical environments** with configurable layouts and social contexts.
- Established evaluation quality through **human-annotated baselines** and controlled scenario augmentation, separating perception failures from privacy reasoning for **attribution analysis**.
- Benchmarked **12 VLMs** and exposed a **gap between privacy knowledge and behavior**: even the best model jointly satisfied task and privacy requirements in only **51%** of conflicting-command cases.

Beyond Steering Vector: Flow-based Activation Steering Mar 2026 - May 2026
Georgia Institute of Technology | First Author | **NeurIPS 2026** |  Website Advisor: Dr. Chao Zhang

- Introduced FLAS, a **flow-based activation steering method** that learns concept-conditioned velocity fields to adapt interventions to each token's evolving activation state.
- Revealed **curved, multi-step, token-dependent steering** trajectories through activation-space analysis, providing empirical evidence that challenges the **linear representation hypothesis** in activation steering.
- Achieved **zero-shot generalization** to unseen concepts without per-concept tuning, becoming the first learned steering method to consistently outperform prompting on AxBench across Gemma-2-2B/9B.

SEAD: A State-Based Perspective on Attack and Defense in Tool-Using Agents Jul 2026 - Sept 2026
Georgia Institute of Technology | First Author |  Website Advisor: Dr. Pan Li

- Formulated agent attack and defense as a **POMDP** and implemented an **agent safety testing sandbox** with **resettable environments, trajectory replay, and executable checks** across filesystem, terminal, database, and web tasks.
- Developed DART, an **evolving attack agent** that refines strategies through execution feedback and tree search, using a **27B model** to attack commercial agents and achieving **73.3% attack success** against GPT-5.6.
- Developed SAGE, a **defender agent**, through **agentic post-training**, which actively **investigates environment** for safety evidence and reduces attack success to **at most 4.0%** under various adaptive attacks.

AcFlow: Controlling T2I DiTs via Learned Conditional Activation Flow Jul 2026 - Sept 2026
Georgia Institute of Technology | First Author |  Website

- Developed ACFlow as a **scalable, text-conditioned, continuous-strength control module** for frozen image generators, enabling control toward a **general-purpose controller** for diffusion transformers.
- Scaled concept coverage without per-concept fitting: one controller covers **15,000+ styles** or **1,000+ concepts** and **generalizes** to unseen concepts.
- Validated the control interface on **FLUX and Z-Image**, achieving the **best intervention-preservation trade-off** among evaluated baselines.

WORK EXPERIENCE

Machine Learning Engineer Intern Sept 2024 - May 2025
Infplane AI Ltd. | Pre-Series A | approx. CNY 50M raised

- Set up and ran small-scale LLM **pretraining experiments** with **Megatron-LM**.
- Prepared pretraining corpora using OpenWebText, including **text cleaning, language/length filtering, and LSH-based deduplication**, followed by tokenization and conversion to indexed binary datasets.

PUBLICATIONS

** denotes equal contribution.*

[1] **How far are VLMs from privacy awareness in the physical world? an empirical study**
Junran Wang*, Xinjie Shen*, Zehao Jin*, and Pan Li. **NeurIPS 2026** [Paper].

[2] **Beyond Steering Vector: Flow-based Activation Steering for Inference-Time Intervention**
Zehao Jin*, Ruixuan Deng*, **Junran Wang***, Xinjie Shen, and Chao Zhang. **NeurIPS 2026** [Paper].

[3] **Accelerated Sequential Flow Matching: A Bayesian Filtering Perspective**
Yinan Huang, Hans Hao-Hsun Hsu, **Junran Wang**, Bo Dai, and Pan Li. **ReALM-GEN 2026 - ICLR 2026 Workshop (Spotlight)** [Paper].

[4] **SEAD: A State-Based Perspective on Attack and Defense in Tool-Using Agents**
Junran Wang*, Xinjie Shen*, Rongzhe Wei, and Pan Li. *arXiv preprint (2026)* [Paper].

[5] **AcFlow: Controlling Text-to-Image Diffusion Transformers via Learned Conditional Activation Flow**
Junran Wang*, Zehao Jin*, Tianyu Luan, and Xinjie Shen. *arXiv preprint (2026)* [Paper].

[6] **Transformers Stop Thinking Too Early, and a Tiny LoRA Fixes It**
Zehao Jin, Ruixuan Deng, and **Junran Wang**. *arXiv preprint (2026)* [Paper].

HONORS & AWARDS

National Scholarship	Top 2% university students in China	2024
Overall Excellence Scholarship	Top 5% in Tsinghua University	2023
Finalist in Interdisciplinary Contest in Modeling (ICM)	Top 2% teams worldwide	2023
Overall Excellence Scholarship	Top 5% in Tsinghua University	2022

SKILLS

Programming Languages	C, C++, Python, familiar with RUST
ML Frameworks & Serving	PyTorch, CUDA, Hugging Face Transformers, vLLM, SGLang
AI Agent Infra	Post-training, MCP, MCP, Trajectory Analysis
Software Engineering	Docker, Docker Compose, Slurm, Git
Rendering	Unity Engine, Blender, Meshlab
Languages	Mandarin (native), English (TOEFL 115), Italian (entry-level)